

Calculus III Final Exam Cheatsheet

Directional derivative - Absolute extrema/Lagrange - Multiple integrals - Polar/Spherical - Scalar line integral - Green's theorem - Flux

Fast recognition. Direction/change $\rightarrow$ directional derivative. Absolute max/min on closed $D \rightarrow$ interior + boundary + corners. Constraint $g = c \rightarrow$ Lagrange. Circles/disks $\rightarrow$ polar. Spheres/cones $\rightarrow$ spherical. $\int_C f ds \rightarrow$ scalar line integral. $\oint_C P dx + Q dy$ with closed $C \rightarrow$ Green. $\iint_S \mathbf{F} \cdot \mathbf{n} dS \rightarrow$ flux.

1. Directional Derivative

For $f(x, y, z)$,

$$\nabla f = \langle f_x, f_y, f_z \rangle$$

For a given direction vector $\mathbf{v}$,

$$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|}$$

Then

$$D_{\mathbf{u}}f(P) = \nabla f(P) \cdot \mathbf{u}$$

If direction is from A to B :

$$\mathbf{v} = B - A, \quad \mathbf{u} = \frac{B - A}{|B - A|}$$

If a direction angle θ is given in the plane:

$$\mathbf{u} = \langle \cos \theta, \sin \theta \rangle$$

Fastest increase:

$$\text{direction} = \frac{\nabla f}{|\nabla f|}, \quad \text{rate} = |\nabla f|$$

Fastest decrease:

$$\text{direction} = -\frac{\nabla f}{|\nabla f|}, \quad \text{rate} = -|\nabla f|$$

Rule of thumb: if the problem gives a raw direction vector, *normalize first*. If $\mathbf{u} \perp \nabla f$, then $D_{\mathbf{u}}f = 0$.

2. Absolute Max/Min + Lagrange

For a continuous f on a closed, bounded region D , absolute extrema exist.

Interior candidates:

$$f_x = 0, \quad f_y = 0$$

Keep only solutions inside D .

Boundary constraint: if $g(x, y) = c$,

$$\nabla f = \lambda \nabla g$$

that is,

$$f_x = \lambda g_x, \quad f_y = \lambda g_y, \quad g = c$$

In 3D add $f_z = \lambda g_z$.

Final comparison checklist:

1. Interior critical points.
2. Every smooth boundary piece (Lagrange or parametrization).

3. Corners/endpoints.

4. Evaluate f at all candidates.

5. Largest value = absolute max; smallest = absolute min.

Trap: Lagrange handles the constrained boundary; it does *not* replace checking the interior.

3. Double and Triple Integrals

General double integral:

$$\iint_D f(x, y) dA$$

Type I region:

$$\iint_D f dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

Area:

$$\iint_D 1 dA = \text{Area}(D)$$

If $f \geq 0$, $\iint_D f dA$ is volume under $z = f(x, y)$.

Triple integral:

$$\iiint_E f(x, y, z) dV$$

Volume:

$$\iiint_E 1 dV = \text{Vol}(E)$$

If $z_{\min}(x, y) \leq z \leq z_{\max}(x, y)$:

$$\iiint_E f dV = \iint_D \int_{z_{\min}}^{z_{\max}} f dz dA$$

Rule: read iterated integrals from inside out.

4. Polar Coordinates

Use for disks, circles, annuli, sectors, or $x^2 + y^2$.

$$x = r \cos \theta, \quad y = r \sin \theta, \quad x^2 + y^2 = r^2$$

$$dA = r dr d\theta$$

Therefore

$$\iint_D f dA = \int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} f(r \cos \theta, r \sin \theta) r dr d\theta$$

Standard angles:

- Full circle: $0 \leq \theta \leq 2\pi$.

- Upper half: $0 \leq \theta \leq \pi$.
- First quadrant: $0 \leq \theta \leq \pi/2$.
- Right half: $-\pi/2 \leq \theta \leq \pi/2$.

Useful conversions:

$$x^2 + y^2 = R^2 \Rightarrow r = R.$$

For other boundaries, substitute $x = r \cos \theta$ and $y = r \sin \theta$ and solve directly for r . **Trap:** never forget the Jacobian factor r .

5. Spherical Coordinates

Use for spheres, hemispheres, cones with vertex at the origin, radial functions.

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$x^2 + y^2 + z^2 = \rho^2, \quad \sqrt{x^2 + y^2} = \rho \sin \phi$$

Volume element:

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

Meaning:

- ρ : distance from origin.
- θ : rotation around the z -axis.
- ϕ : angle measured down from the positive z -axis.

Whole sphere: $0 \leq \rho \leq R$, $0 \leq \phi \leq \pi$, $0 \leq \theta \leq 2\pi$. Upper hemisphere: $0 \leq \phi \leq \pi/2$.

Cone trick: for $z = \sqrt{x^2 + y^2}$,

$$\rho \cos \phi = \rho \sin \phi \Rightarrow \phi = \frac{\pi}{4}.$$

Memory rule: sphere $\rightarrow \rho$ bound; cone $\rightarrow \phi$ bound; rotation $\rightarrow \theta$ bound.

6. Scalar Line Integral (Type 1)

Parametrize the curve:

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$$

Then

$$\mathbf{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle$$

$$ds = |\mathbf{r}'(t)| \, dt$$

where

$$|\mathbf{r}'(t)| = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}$$

Core formula:

$$\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| \, dt$$

In 2D simply omit $z'(t)^2$.

Special case:

$$\int_C 1 \, ds = \text{length of } C$$

If f is linear density, $\int_C f \, ds$ is mass of a wire.

Orientation:

$$\int_C f \, ds = \int_{-C} f \, ds$$

Type 1 does not care which way the curve is traversed.

Useful parametrizations:

$$\mathbf{r}(t) = A + t(B - A), \quad 0 \leq t \leq 1$$

for a line segment, and

$$x = a + R \cos t, \quad y = b + R \sin t$$

for a circle.

7. Green's Theorem

For $\mathbf{F} = (P, Q)$ and positively oriented (counterclockwise) closed curve $C = \partial D$:

$$\oint_C P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dA$$

Memorize:

$$Q_x - P_y$$

Positive orientation = counterclockwise = region stays on your left.

If C is clockwise:

$$\oint_C P \, dx + Q \, dy = - \iint_D (Q_x - P_y) \, dA$$

Workflow: identify $P, Q \rightarrow$ compute $Q_x - P_y \rightarrow$ find region $D \rightarrow$ integrate.

Trap: Green applies to a *vector/type-2 line integral*, not to $\int_C f \, ds$.

8. Surface Integral Type 2 (Flux)

For vector field

$$\mathbf{F} = (P, Q, R)$$

flux through an oriented surface S is

$$I = \iint_S \mathbf{F} \cdot \mathbf{n} \, dS$$

Lecture notation:

$$I = \iint_S P \, dy \, dz + Q \, dz \, dx + R \, dx \, dy$$

$\mathbf{F} \cdot \mathbf{n}$ extracts the component of the field normal to the surface.

If a surface is implicit, $G(x, y, z) = 0$:

$$\nabla G = \langle G_x, G_y, G_z \rangle \text{ is normal to } S$$

Unit normal:

$$\mathbf{n} = \pm \frac{\nabla G}{\|\nabla G\|}$$

Graph shortcut: $z = g(x, y)$

For **upward** orientation:

$$\mathbf{n} dS = \langle -g_x, -g_y, 1 \rangle dx dy$$

Hence

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_D (-Pg_x - Qg_y + R) dx dy$$

For **downward** orientation:

$$\mathbf{n} dS = \langle g_x, g_y, -1 \rangle dx dy$$

Hence

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_D (Pg_x + Qg_y - R) dx dy$$

Evaluate P, Q, R on the surface $z = g(x, y)$.

Why no normalization? In the product $\mathbf{n} dS$, the unit-normal denominator $\sqrt{1 + g_x^2 + g_y^2}$ cancels the same factor in dS .

Parametric surface

If $\mathbf{r}(u, v)$ parametrizes S ,

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) du dv$$

If orientation is opposite, multiply the normal by -1 .

9. Physical / Geometric Meaning

- $\int_a^b f(x) dx$: accumulation along an interval.

- $\iint_D f dA$: accumulation over area; if $f \geq 0$, volume under a surface.
- $\iiint_E f dV$: accumulation over volume; $f = 1$ gives volume.
- $\int_C f ds$: accumulation along curve length; $f = 1$ gives curve length.
- $\iint_S \mathbf{F} \cdot \mathbf{n} dS$: net normal flow (flux) through a surface.

Flux signs (for chosen normal):

- $\Phi > 0$: net flow in the normal direction.
- $\Phi < 0$: net flow opposite the normal direction.
- $\Phi = 0$: tangent flow or cancellation of positive/negative contributions; field need not be zero.

10. Last-Minute Traps

1. Directional derivative: normalize the direction vector.
2. Absolute extrema: check interior + boundary + corners.
3. Polar: $dA = r dr d\theta$.
4. Spherical: $dV = \rho^2 \sin \phi d\rho d\phi d\theta$.
5. θ rotates around z ; ϕ is measured down from $+z$.
6. Scalar line integral: include $ds = |\mathbf{r}'(t)|dt$.
7. Green: curve must be closed; CCW is positive.
8. Flux: orientation matters; reversing the normal changes the sign.
9. Graph-flux shortcut already combines $\mathbf{n}$ and dS ; do not normalize again.

10 formulas to memorize first: ∇f , $\mathbf{u} = \mathbf{v}/|\mathbf{v}|$, $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$, $\nabla f = \lambda \nabla g$, $dA = r dr d\theta$, $dV = \rho^2 \sin \phi d\rho d\phi d\theta$, $\int_C f ds = \int f(\mathbf{r}(t))|\mathbf{r}'(t)|dt$, $\oint_C Pdx + Qdy = \iint_D (Q_x - P_y)dA$, $I = \iint_S \mathbf{F} \cdot \mathbf{n} dS$, $z = g(x, y)$: up $(-g_x, -g_y, 1)$, down $(g_x, g_y, -1)$.